A Cash-Allowed Robust Forecast-Aware Portfolio Optimization Framework for Multi-Asset ETF Allocation

Junyuan Ding

Abstract

This paper develops a cash-allowed robust forecast-aware portfolio optimization framework for multi-asset ETF allocation. Traditional mean-variance optimization remains a foundational approach to portfolio construction, but its practical performance is often fragile because expected returns and covariances are difficult to estimate. The research question is whether interpretable forecast signals and convex risk controls can improve allocation behavior relative to standard portfolio rules. The proposed model constructs expected-return forecasts from 12-1 momentum and 200-day trend signals, estimates risk with annualized 20-day EWMA covariance and positive-semidefinite repair, incorporates a robust penalty for forecast uncertainty, includes a BIL cash sleeve, controls implementation cost through full-vector turnover, and imposes a dynamic target-risk budget. Monthly rolling out-of-sample backtests compare the proposed strategy with Equal Weight, Minimum Variance, and Traditional MVO baselines. In the tested sample, the proposed strategy achieves a 9.8% annualized net return, 10.2% annualized volatility, a Sharpe ratio of 0.96, and a maximum drawdown of -15.7%, compared with Sharpe ratios of 0.75, 0.73, and 0.79 for Equal Weight, Minimum Variance, and Traditional MVO. The results suggest that robust forecast-aware optimization with an explicit cash sleeve can improve certain risk-adjusted metrics in this ETF universe, although performance remains dependent on forecast quality, volatility estimation, transaction-cost assumptions, and market regimes.

1 Introduction

Asset allocation is a central problem for investors who allocate capital across equities, government bonds, credit, real assets, and cash-like instruments. Exchange traded funds make this problem especially relevant because they provide liquid and diversified exposure to broad asset classes. Unlike single-stock selection, a multi-asset ETF allocation problem naturally emphasizes portfolio-level properties: return, volatility, drawdown, turnover, and implementation cost. A useful allocation framework should therefore be evaluated not only by cumulative return, but also by risk-adjusted performance, exposure dynamics, and trading frictions.

Markowitz mean-variance optimization provides the classical framework for balancing expected return and risk (Markowitz, 1952). Its practical implementation, however, is sensitive to estimation error. Expected returns are especially noisy, finite-sample covariance matrices may be unstable, and optimized weights can become extreme when the optimizer exploits estimation noise. A fully invested risky-asset model is also unable to reduce aggregate risky exposure during market stress unless cash is explicitly included as an allocation choice. These issues help explain why simple allocation rules such as equal weighting remain important benchmarks in empirical portfolio research.

This paper studies a forecast-aware but conservative alternative. Price-based momentum and trend variables may contain useful information about relative expected returns across asset classes, but their predictive ability is imperfect and regime-dependent. The proposed strategy therefore does not use momentum as a standalone ranking rule. Instead, momentum and trend signals are converted into an annualized alpha vector that enters a convex robust optimization problem. The optimizer balances forecasted alpha against EWMA risk, robust forecast uncertainty, concentration regularization, transaction costs, and a dynamic target-risk budget.

The proposed model allocates across a risky ETF universe and a BIL cash sleeve. The risky ETF universe includes equity, Treasury, credit, and real asset ETFs, while BIL is used only as a cash proxy. The final strategy uses annualized 20-day EWMA covariance estimation with positive-semidefinite repair, a volatility-adaptive robust penalty, a dynamic risk budget based on recent market stress, full-vector turnover costs, and monthly rolling out-of-sample backtesting. The model remains a convex optimization problem and is implemented with a second-order-cone-compatible risk term.

The paper makes three contributions. First, it integrates medium-term momentum and trend forecasts into a robust convex optimization framework for multi-asset ETF allocation. Second, it allows cash allocation through an explicit BIL sleeve, so the optimizer is not forced to remain fully invested in risky assets. Third, it evaluates the strategy using cumulative performance, drawdown, Sharpe and Sortino ratios, cash and group exposure, signal diagnostics, volatility forecast diagnostics, turnover, transaction-cost drag, and parameter sensitivity tests. The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 describes the ETF universe and data construction. Section 4 presents the methodology. Section 5 defines the baselines. Sections 6 through 8 report the empirical results, diagnostics, and robustness analysis. Section 10 discusses limitations, and Section 11 concludes.

2 Literature Review

2.1 Mean-Variance Optimization and Estimation Error

The starting point is the mean-variance framework of Markowitz (1952), which formalizes portfolio selection as a tradeoff between expected return and variance. The framework is theoretically elegant, but its empirical performance depends on estimated inputs. DeMiguel et al. (2009) show that naive $1/N$ diversification is a difficult benchmark to beat after accounting for estimation error. This evidence motivates the inclusion of Equal Weight as a primary baseline. Covariance estimation error is another important source of instability. Ledoit and Wolf (2004) propose covariance shrinkage to improve conditioning, while Jagannathan and Ma (2003) show that portfolio constraints such as no-short-sale restrictions can reduce effective estimation error and act as implicit regularization. In the present paper, Ledoit-Wolf shrinkage is discussed as background and as a covariance sensitivity case; the final implemented proposed strategy uses EWMA covariance estimation with positive-semidefinite repair.

2.2 Robust and Convex Portfolio Optimization

Robust portfolio optimization addresses parameter uncertainty by optimizing against adverse perturbations in model inputs. Goldfarb and Iyengar (2003) develop robust portfolio selection under uncertainty sets for returns and covariances. The robust penalty in this paper follows the same broad logic: the alpha forecast is useful but uncertain, so the objective subtracts a worst-case return penalty scaled by portfolio risk. The implementation also follows the practical convex trading perspective of Boyd et al. (2017), where turnover, transaction costs, and implementability are modeled directly rather than treated as afterthoughts.

2.3 Momentum, Trend, and Tactical Allocation

Momentum and trend signals have been documented across asset classes. Moskowitz et al. (2012) study time-series momentum in futures markets, while Asness et al. (2013) examine value and momentum across markets and asset classes. Faber (2007) provides an early tactical asset allocation approach based on moving-average signals. In this paper, such signals are not used as isolated trading rules. They form a forecast layer that feeds a constrained optimization model, and the optimizer is allowed to reduce risky exposure when the forecast-risk tradeoff is unattractive.

2.4 Volatility Targeting and Risk Management

Volatility-managed strategies reduce exposure when estimated risk is high. Moreira and Muir (2017), Barroso and Santa-Clara (2015), and Harvey et al. (2018) show that volatility scaling can materially affect risk-adjusted returns in several contexts. The present model uses this idea through a dynamic risk budget, but the target volatility is not set equal to recent realized volatility. Instead, it is a risk budget adjusted by a multi-asset market stress ratio. This distinction is important: the optimizer still solves a portfolio problem under constraints rather than simply scaling a fixed return stream.

3 Data and Asset Universe

The empirical analysis uses Yahoo Finance adjusted close prices for 14 ETFs from January 4, 2010 through January 30, 2026. After return construction and warm-up, the out-of-sample backtest return sample runs from February 1, 2011 through January 30, 2026. Daily simple returns are computed from adjusted close prices. Portfolios are rebalanced monthly: at each month-end trading day, the model uses only information available up to that date to estimate signals and risks; the resulting weights are applied from the next trading day until the next rebalance date.

Table 1: ETF universe and modeling roles. BIL is used only as the cash proxy.

Group	Tickers	Role	Alpha	Covariance	Cash Sleeve
Equity	SPY, QQQ, IWM, EFA, EEM	Risky ETF	Yes	Yes	No
Treasury	TLT, IEF, SHY	Risky ETF	Yes	Yes	No
Credit	LQD, HYG	Risky ETF	Yes	Yes	No
Real Assets	GLD, DBC, VNQ	Risky ETF	Yes	Yes	No
Cash	BIL	Cash proxy	No	No	Yes

Table 1 summarizes the ETF universe. The risky universe contains five equity ETFs, three Treasury ETFs, two credit ETFs, and three real asset ETFs. BIL is treated as a cash proxy. It is excluded from alpha signal construction and risky covariance estimation, and it enters only through the cash sleeve return in the proposed strategy. The three baseline strategies are risky-only strategies; for turnover accounting their full portfolio vector is represented as $[x, 0]$, where the last component is the zero BIL weight.

4 Methodology

Let N be the number of risky ETFs. At rebalance date t, let $x_t \in \mathbb{R}^N$ denote risky ETF weights and c_t denote the cash weight. Let $\hat{\alpha}_t$ be the annualized forecast vector and $\hat{\Sigma}_t$ be the estimated annualized risky covariance matrix. The pre-trade weights after market drift are denoted by x_t^{pre} and c_t^{pre}.

4.1 Forecast Signal Construction

The alpha signal is constructed only for risky ETFs. For risky ETF i, the 12-1 momentum signal is

$$m_{i,t} = \frac{P_{i,t-21}}{P_{i,t-252}} - 1. \tag{1}$$

This uses approximately one year of price history while skipping the most recent 21 trading days. The cross-sectional momentum z-score is

$$z_{i,t}^{mom} = \frac{m_{i,t} - \bar{m}_t}{s_t(m)}, \tag{2}$$

where the mean and standard deviation are computed across risky ETFs only. If cross-sectional dispersion is numerically negligible, the implementation returns a zero z-score rather than amplifying noise. The trend signal is

$$trend_{i,t} = \begin{cases} 1, & P_{i,t} > MA_{200,i,t}, \\ -1, & P_{i,t} \le MA_{200,i,t}. \end{cases} \tag{3}$$

The combined score and annualized alpha are

$$score_{i,t} = z_{i,t}^{mom} + 0.5\, trend_{i,t}, \qquad \hat{\alpha}_{i,t} = 0.03\, score_{i,t}. \tag{4}$$

The score should be interpreted as a relative expected-return forecast, not as a literal point forecast of future realized returns.

4.2 EWMA Covariance Estimation with PSD Repair

The final proposed strategy does not use Ledoit-Wolf shrinkage as its implemented risk model. It uses an annualized 20-trading-day EWMA covariance estimator with decay $\lambda = 0.94$. For a covariance window with $n = 20$ cleaned daily return observations $r_{t-n+1}, \dots, r_t$, the implementation first subtracts the window mean $\bar{r}_t$. It then assigns normalized exponentially decaying weights

$$\omega_j = \frac{(1-\lambda)\lambda^{n-j}}{\sum_{\ell=1}^{n}(1-\lambda)\lambda^{n-\ell}}, \qquad j = 1, \dots, n, \tag{5}$$

where $j = n$ denotes the most recent observation. The annualized covariance matrix is

$$\hat{\Sigma}_t = 252 \sum_{j=1}^{n} \omega_j \left(r_{t-n+j} - \bar{r}_t\right)\left(r_{t-n+j} - \bar{r}_t\right)^{\top}. \tag{6}$$

After estimation, the covariance matrix is symmetrized and repaired by eigenvalue clipping to ensure positive semidefiniteness. The optimization uses the repaired matrix.

4.3 Market Stress, Robust Penalty, and Dynamic Risk Budget

The market stress measure is based on the risky equal-weight ETF return, excluding BIL. The numerator is the latest annualized span-20 EWMA volatility of this risky equal-weight return, evaluated over the 252-day stress window, and the denominator is its annualized 252-day realized volatility:

$$v_t = \mathrm{clip}\left(\frac{\sigma_t^{EW,EWMA20}}{\max(\sigma_t^{EW,252}, \epsilon)}, 0.5, 2.5\right). \tag{7}$$

The robust penalty coefficient and target-risk budget are

$$\rho_t = \mathrm{clip}(0.08v_t, 0.03, 0.25), \tag{8}$$

$$\sigma_t^* = \mathrm{clip}\left(\frac{0.10}{v_t}, 0.04, 0.16\right). \tag{9}$$

Thus elevated recent market stress increases the robust penalty and lowers the allowed risk budget. The target σ_t^* is a dynamic risk budget used in the optimization; it is not defined to be equal to recent realized volatility.

4.4 Robust Cash-Allowed Optimization

The proposed strategy solves the following convex optimization problem:

$$\max_{x_t, c_t} \quad \hat{\alpha}_t^\top x_t - \lambda_{tc} TO_t - \eta \|x_t\|_2^2 - \rho_t R_t \tag{10}$$

$$\text{s.t.} \quad \mathbf{1}^\top x_t + c_t = 1, \tag{11}$$

$$x_t \geq 0, \quad c_t \geq 0, \tag{12}$$

$$x_{i,t} \leq 0.25, \quad i = 1, \ldots, N, \tag{13}$$

$$R_t \leq \sigma_t^*, \tag{14}$$

where

$$R_t = \|\hat{\Sigma}_t^{1/2} x_t\|_2 = \sqrt{x_t^\top \hat{\Sigma}_t x_t}. \tag{15}$$

The implementation represents R_t with the SOC norm above. It does not use a square-rooted quadratic-form expression, which helps maintain DCP compliance in CVXPY. Full-vector one-way turnover is

$$TO_t = \frac{1}{2}\left\| \begin{bmatrix} x_t \\ c_t \end{bmatrix} - \begin{bmatrix} x_t^{pre} \\ c_t^{pre} \end{bmatrix} \right\|_1 \tag{16}$$

for the proposed strategy. For risky-only baselines, the full vector is $[x_t, 0]$. The objective uses $\lambda_{tc} = 0.001$ as a mild turnover penalty and $\eta = 0.01$ for L2 concentration regularization. The backtest deducts actual transaction costs $0.001 \times TO_t$ at rebalancing.

4.5 Robust Penalty Interpretation

For exposition, assume $\hat{\Sigma}_t$ is positive definite after numerical repair. Suppose the true alpha is

$$\alpha_t = \hat{\alpha}_t + \delta_t, \tag{17}$$

where δ_t lies in the covariance-scaled uncertainty set

$$\mathcal{U}_t = \{\delta_t : \delta_t^\top \hat{\Sigma}_t^{-1} \delta_t \leq \rho_t^2\}. \tag{18}$$

Then by the Cauchy-Schwarz inequality in the covariance metric,

$$\min_{\delta_t \in \mathcal{U}_t} (\hat{\alpha}_t + \delta_t)^\top x_t = \hat{\alpha}_t^\top x_t - \rho_t \sqrt{x_t^\top \hat{\Sigma}_t x_t}. \tag{19}$$

This derivation motivates the robust penalty as protection against alpha forecast error in directions that are risky under the estimated covariance matrix.

4.6 Convexity and Numerical Solution

The proposed monthly optimization problem is a convex optimization problem in the standard sense: it maximizes a concave objective over a convex feasible set. The forecast term $\hat{\alpha}_t^\top x_t$ is linear. Full-vector turnover TO_t is proportional to an L1 norm and is therefore convex, so $-\lambda_{tc} TO_t$ is concave in the maximization form. The concentration term $\|x_t\|_2^2$ is convex, so $-\eta\|x_t\|_2^2$ is concave. The risk term

$$R_t = \|\hat{\Sigma}_{sqrt,t} x_t\|_2$$

is a convex norm, so $-\rho_t R_t$ is concave. The budget constraint is affine, the long-only and single-asset upper-bound constraints are box constraints, and the volatility constraint $R_t \leq \sigma_t^*$ is a second-order cone constraint.

Equivalently, the problem can be written as a convex minimization problem. Let $\hat{\Sigma}_{sqrt,t}$ be a square-root factor of the PSD-repaired covariance matrix. Introduce a scalar risk variable r_t and an auxiliary nonnegative turnover vector $u_t \in \mathbb{R}_+^{N+1}$. Define

$$w_t = \begin{bmatrix} x_t \\ c_t \end{bmatrix}, \qquad w_t^{pre} = \begin{bmatrix} x_t^{pre} \\ c_t^{pre} \end{bmatrix}. \tag{20}$$

At each rebalance date, $\hat{\alpha}_t$, $\hat{\Sigma}_{sqrt,t}$, ρ_t, σ_t^*, and w_t^{pre} are fixed inputs computed from historical data, while x_t, c_t, r_t, and u_t are decision variables. An equivalent convex minimization form suitable for conic and quadratic-conic solvers is

$$
\begin{aligned}
\min_{x_t, c_t, r_t, u_t} \quad & -\hat{\alpha}_t^\top x_t + \lambda_{tc}\left(\frac{1}{2}\mathbf{1}^\top u_t\right) + \eta\|x_t\|_2^2 + \rho_t r_t & (21)\\
\text{s.t.} \quad & \mathbf{1}^\top x_t + c_t = 1, & (22)\\
& x_t \geq 0, \quad c_t \geq 0, & (23)\\
& x_{i,t} \leq 0.25, \quad i = 1, \dots, N, & (24)\\
& \|\hat{\Sigma}_{sqrt,t} x_t\|_2 \leq r_t, & (25)\\
& r_t \geq 0, & (26)\\
& r_t \leq \sigma_t^*, & (27)\\
& u_t \geq w_t - w_t^{pre}, & (28)\\
& u_t \geq -(w_t - w_t^{pre}), & (29)\\
& u_t \geq 0. & (30)
\end{aligned}
$$

The turnover used in the objective and in transaction-cost accounting is then

$$TO_t = \frac{1}{2}\mathbf{1}^\top u_t. \tag{31}$$

This formulation uses affine constraints, box constraints, a convex quadratic term in the objective, and second-order cone constraints, making it directly compatible with modern conic or quadratic-conic optimization solvers. The paper does not derive closed-form KKT solutions because the monthly optimization includes L1 turnover penalties, SOC risk constraints, a cash sleeve, and time-varying forecasts, covariance matrices, robust penalties, and dynamic risk budgets. Instead, the optimizer is solved numerically at each rebalance date using CVXPY with a conic solver, and the implementation checks solver status, feasibility, budget balance, upper-bound violations, and target-volatility violations after each optimization.

4.7 Performance Metrics

All performance metrics are computed from daily net returns unless stated otherwise. Annualized return is the geometric annualized return $(\prod_t(1+r_t))^{252/T} - 1$. Annualized volatility is the daily return sample standard deviation multiplied by $\sqrt{252}$. The Sharpe ratio uses a zero annual risk-free rate and is computed as daily excess-return mean divided by daily excess-return sample standard deviation, annualized by $\sqrt{252}$. The Sortino ratio uses target return zero and downside return sample standard deviation. Maximum drawdown is the minimum value of cumulative wealth divided by its running maximum minus one. The Calmar ratio is annualized return divided by the absolute maximum drawdown. Transaction-cost drag in the main table is the annualized gross return minus the annualized net return.

5 Baseline Strategies

The final empirical comparison includes exactly four strategies. Equal Weight allocates equally across the 13 risky ETFs and does not use BIL. Minimum Variance solves

$$\min_x x^\top \hat{\Sigma}_t x \quad \text{s.t.} \quad \mathbf{1}^\top x = 1, \quad x \geq 0, \quad x_i \leq 0.25, \tag{32}$$

using a 252-day sample covariance matrix. Traditional MVO solves

$$\max_x \hat{\mu}_t^\top x - \gamma x^\top \hat{\Sigma}_t x \quad \text{s.t.} \quad \mathbf{1}^\top x = 1, \quad x \geq 0, \quad x_i \leq 0.25, \tag{33}$$

where $\hat{\mu}_t$ is the daily mean return over the strategy's covariance window multiplied by 252, $\hat{\Sigma}_t$ is the corresponding covariance estimate, and $\gamma = 1.0$. In the default implementation, Traditional MVO uses a 252-day sample covariance matrix. The Proposed Strategy is the full cash-allowed robust forecast-aware model described in Section 4. It differs from Traditional MVO by using signal-based alpha forecasts, a cash sleeve, EWMA covariance with PSD repair, a robust forecast-uncertainty penalty, a dynamic risk budget, L2 regularization, and explicit full-vector turnover costs.

6 Empirical Results

6.1 Main Performance

Table 2: Main performance summary using net returns after transaction costs.

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar	Final NV	Avg TO	Cost Drag	Avg Cash
Equal Weight	6.9%	9.5%	0.75	-20.7%	0.34	2.73	1.7%	0.02%	0.0%
Minimum Variance	3.1%	4.3%	0.73	-14.6%	0.21	1.59	3.4%	0.04%	0.0%
Traditional MVO	9.2%	12.1%	0.79	-19.6%	0.47	3.74	19.0%	0.25%	0.0%
Proposed Strategy	9.8%	10.2%	0.96	-15.7%	0.62	4.04	27.7%	0.37%	9.8%

Table 2 reports the main out-of-sample results after transaction costs. In the tested sample, the proposed strategy has the highest final net value among the four strategies and the highest Sharpe ratio. Its annualized net return is 9.8%, compared with 6.9% for Equal Weight, 3.1% for Minimum Variance, and 9.2% for Traditional MVO. Its annualized volatility is lower than Traditional MVO but higher than Minimum Variance. The proposed strategy also has a smaller maximum drawdown than Equal Weight and Traditional MVO, although it does not have the lowest drawdown overall; the Minimum Variance strategy has the smallest maximum drawdown but substantially lower return. These results suggest an improvement in risk-adjusted performance relative to the selected baselines, not a guarantee of dominance in all market environments.

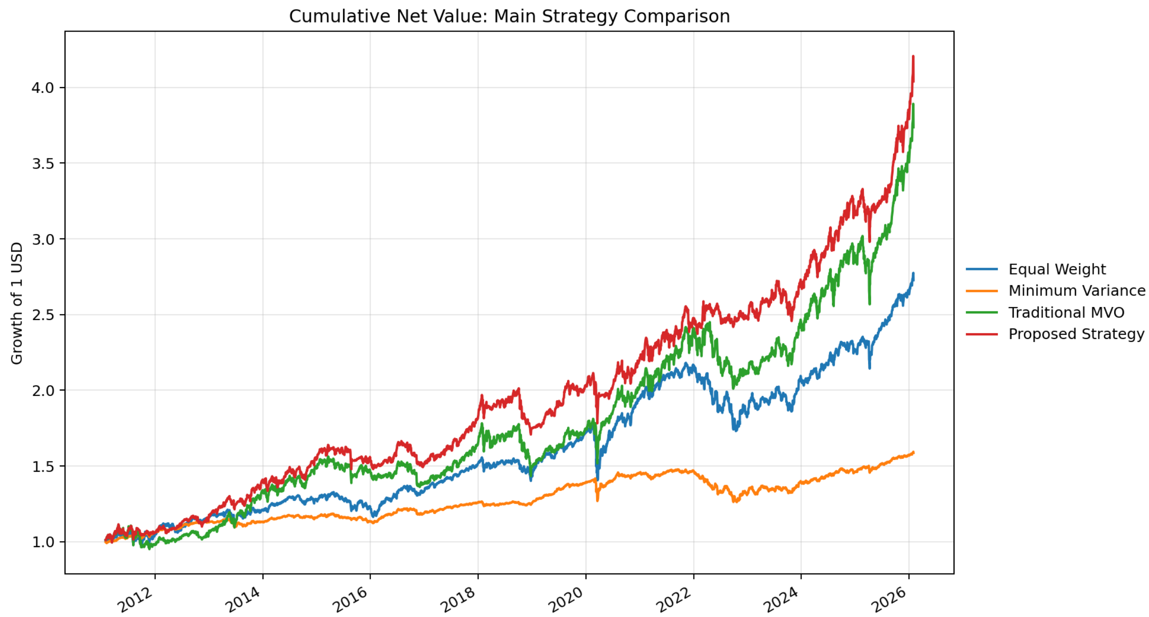

Figure 1: Cumulative net value for the four final strategies. The figure uses daily net returns after transaction costs and contains only Equal Weight, Minimum Variance, Traditional MVO, and the Proposed Strategy.

Figure 1 shows the cumulative net value path. The proposed strategy compounds more strongly over the full sample, while Traditional MVO also achieves high terminal wealth but with higher volatility and a larger drawdown. The comparison should be interpreted jointly with risk and turnover metrics rather than as a cumulative-return ranking alone.

6.2 Drawdown Behavior

Table 3: Maximum drawdown periods for the four main strategies.

Strategy	Max DD	Start	Valley	Recovery	Days
Equal Weight	-20.7%	2021-11-09	2022-10-14	2024-06-18	952
Minimum Variance	-14.6%	2021-09-15	2022-10-20	2024-09-19	1100
Traditional MVO	-19.6%	2018-01-26	2018-12-24	2020-01-16	720
Proposed Strategy	-15.7%	2020-02-19	2020-03-19	2020-07-29	161

Table 3 and Figure 2 summarize drawdown behavior. The proposed strategy's maximum drawdown is -15.7%, occurring from February 19, 2020 to March 19, 2020 and recovering by July 29, 2020. Equal Weight and Traditional MVO experience maximum drawdowns of -20.7% and -19.6%, respectively. Minimum Variance has a smaller maximum drawdown of -14.6%, but its drawdown period lasts substantially longer. The results indicate that the proposed strategy reduces drawdown relative to the more return-seeking baselines, while still remaining exposed to sudden stress periods.

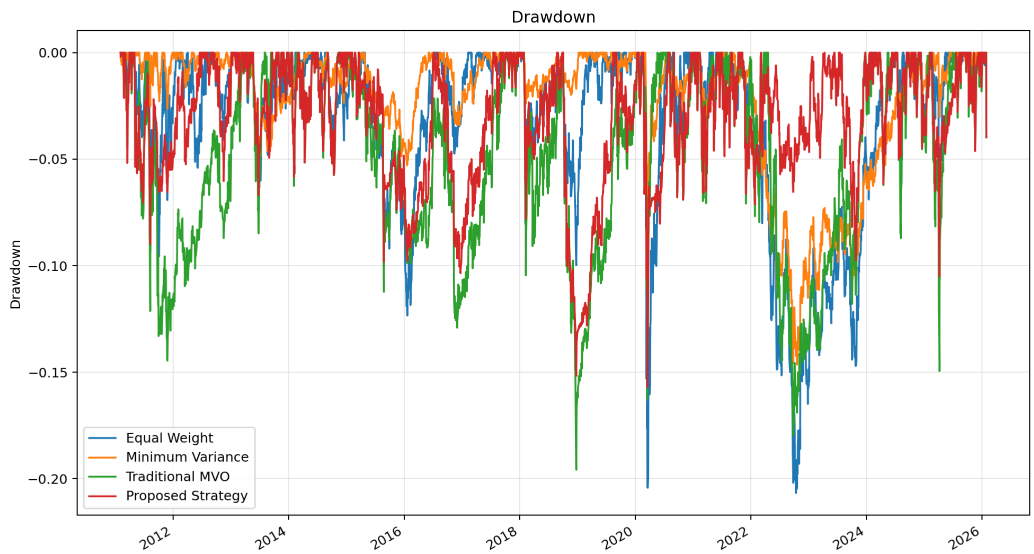

Figure 2: Drawdown curves for the four final strategies. The proposed strategy reduces drawdown relative to Equal Weight and Traditional MVO in the tested sample but does not eliminate drawdown risk.

7 Diagnostic Analysis

7.1 Cash and Exposure Dynamics

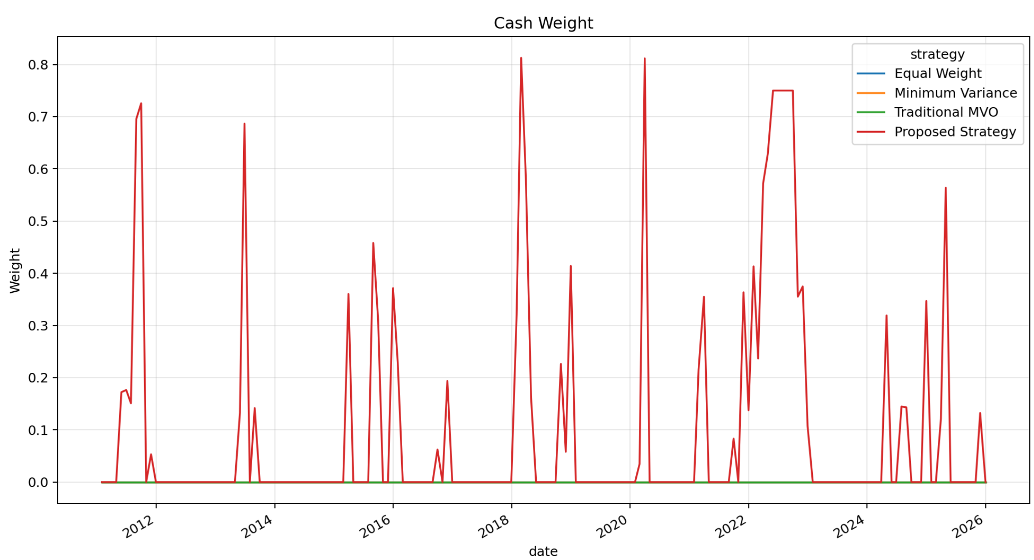

Figure 3: Cash weight of the proposed strategy. The BIL sleeve allows the optimizer to reduce risky exposure when the forecast-risk tradeoff is unattractive.

Figure 3 shows that the proposed strategy uses the cash sleeve meaningfully. The average cash weight is 9.8%, and the maximum cash weight reaches 81.3% in the generated exposure summary. This does not mean the model is a pure market-timing rule; rather, cash is one feasible allocation choice within a constrained optimization problem.

Figure 4 illustrates that the proposed strategy shifts exposure across asset groups over time.

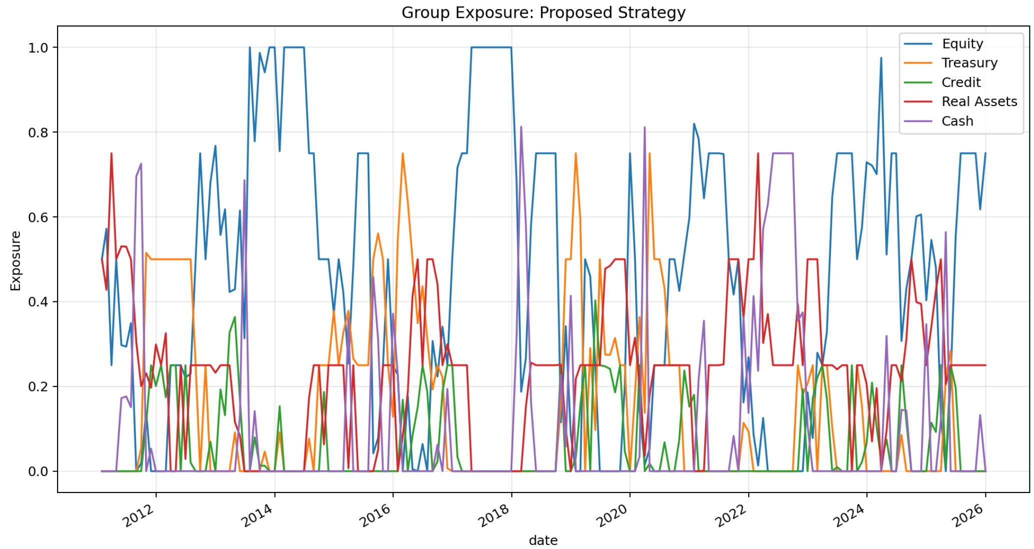

Figure 4: Group exposures of the proposed strategy. Exposures vary across equity, Treasury, credit, real assets, and cash through the rolling optimization.

The average risky exposure is 90.2%, with average equity exposure of 46.1%, Treasury exposure of 14.5%, credit exposure of 6.3%, and real asset exposure of 23.2%. During the proposed strategy's maximum drawdown window, cash exposure is modest, which helps explain why the model still experiences a meaningful drawdown in the 2020 crash.

7.2 Risk Forecast Diagnostics

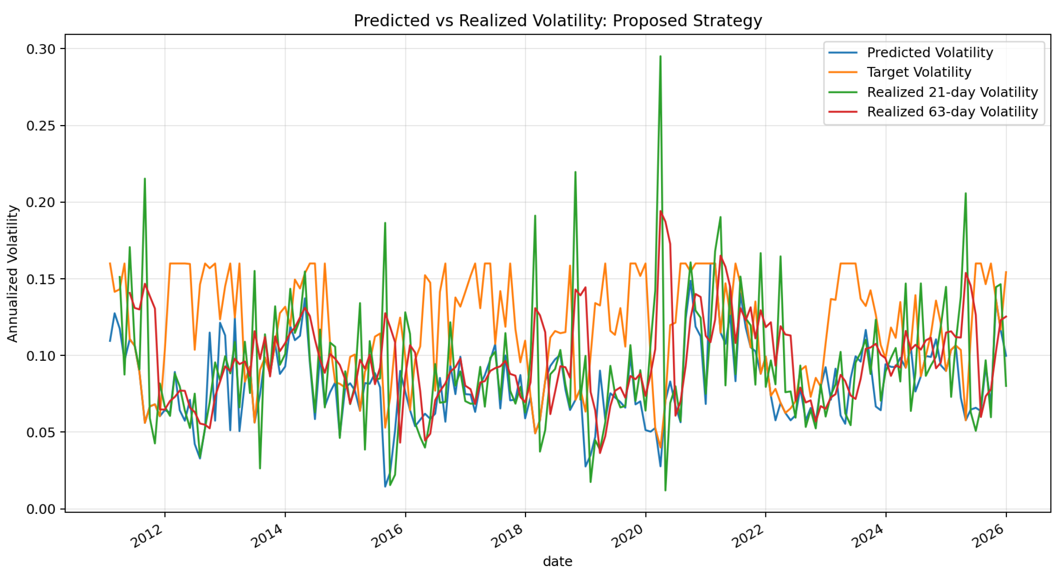

Figure 5: Predicted volatility, dynamic target volatility, and realized volatility for the proposed strategy. Realized volatility can exceed predicted volatility during stress periods.

Table 4: Signal and risk-forecast diagnostics for the proposed strategy.

Diagnostic	Value
Average cross-sectional IC	0.072
IC standard deviation	0.458
IC t-statistic	2.11
IC during normal periods	0.088
IC during drawdown periods	0.015
Positive IC ratio	55.3%
Average predicted volatility	8.2%
Average target volatility	11.8%
Average realized 21-day volatility	9.3%
Average realized/predicted ratio	1.28
Target-volatility binding rate	28.3%

The risk diagnostics in Table 4 and Figure 5 show that the model's average predicted volatility is 8.2%, while average realized 21-day volatility is 9.3% and average realized 63-day volatility is 9.8%. The average realized-to-predicted volatility ratio is 1.28, and the maximum ratio is much larger during stress. This indicates that the EWMA risk model is responsive but can still underestimate future realized risk in abrupt market moves. The target-volatility constraint binds in 28.3% of rebalance periods, so the risk budget is active but not always the dominant constraint.

7.3 Signal Diagnostics

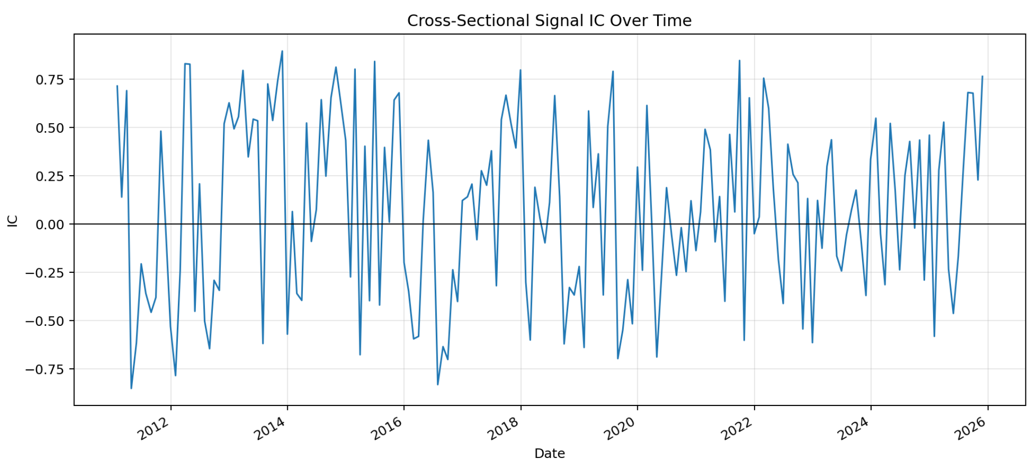

Figure 6: Monthly cross-sectional information coefficient of the forecast signal. The signal is positive on average but noisy across rebalance periods.

The average cross-sectional information coefficient is 0.072 with a t-statistic of 2.11 and a positive IC ratio of 55.3%. The signal is stronger in normal periods than during drawdowns: average IC is 0.088 in normal periods and 0.015 in drawdown periods. These diagnostics support a cautious interpretation. The momentum and trend signal appears informative on average in the tested sample, but its predictive power is far from stable, particularly during adverse market regimes.

7.4 Turnover and Transaction Costs

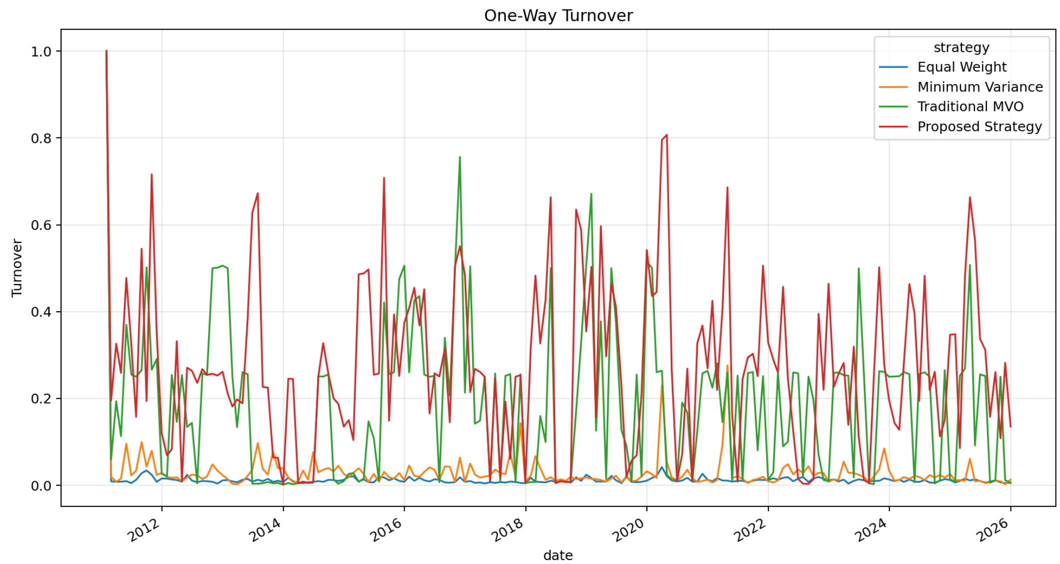

Figure 7: One-way turnover over time. Turnover is computed using the full vector, including the BIL cash sleeve for the proposed strategy.

The proposed strategy has average one-way turnover of 27.7%, higher than the three baselines. At a 10 bps one-way transaction cost rate, its annualized transaction-cost drag is 0.37%, and its final gross value of 4.25 is reduced to a final net value of 4.04. Transaction costs therefore matter, but they do not fully explain the difference between the proposed strategy and the baselines in this sample.

8 Robustness Analysis

The sensitivity tests are used to examine parameter dependence rather than to select parameters in sample. The final proposed strategy remains the fixed specification defined in Section 4.

Table 5: Target-volatility sensitivity for the proposed strategy.

Target Volatility	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar	Avg Cash	Avg TO
6.0%	8.5%	8.6%	0.99	-13.4%	0.63	18.7%	35.0%
8.0%	9.5%	9.6%	0.98	-14.2%	0.66	12.9%	31.1%
10.0%	9.8%	10.2%	0.96	-15.7%	0.62	9.8%	27.7%
12.0%	9.6%	10.6%	0.92	-16.5%	0.58	8.1%	26.3%

Table 5 shows that lower base target-volatility settings increase average cash weight and reduce drawdown, while higher settings increase risky exposure. The 10% setting used in the final model is not selected by ex post performance maximization; it is the fixed specification used for the main backtest.

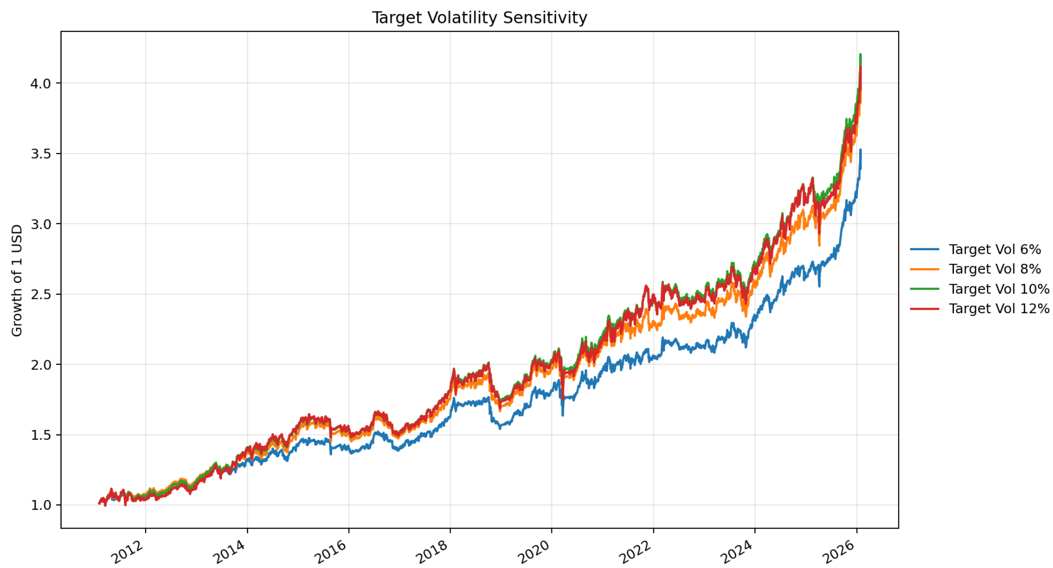

Figure 8: Target-volatility sensitivity for the proposed strategy. The grid changes the base risk budget while preserving the same signal and robust optimization structure.

Table 6: Alpha-scale sensitivity for the proposed strategy.

Alpha Scale	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar	Avg Cash	Avg TO
1.0%	9.3%	9.3%	1.00	-13.8%	0.68	15.9%	27.1%
2.0%	9.4%	10.0%	0.95	-15.1%	0.63	11.0%	27.5%
3.0%	9.8%	10.2%	0.96	-15.7%	0.62	9.8%	27.7%
5.0%	10.1%	10.4%	0.98	-17.0%	0.59	9.0%	28.2%

Table 6 varies the alpha scale. Larger alpha scale generally increases return but also increases drawdown and slightly reduces cash allocation. This is consistent with the interpretation that the alpha signal is useful but noisy: giving it too much weight can increase exposure to forecast error.

Table 7: Robust penalty sensitivity for the proposed strategy.

Rho Setting	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar	Avg Cash
rho_multiplier=0	10.2%	10.5%	0.98	-19.3%	0.53	8.9%
rho_multiplier=0.5	10.2%	10.4%	0.98	-17.5%	0.58	9.2%
rho_multiplier=1	9.8%	10.2%	0.96	-15.7%	0.62	9.8%
rho_multiplier=1.5	9.5%	10.1%	0.95	-14.7%	0.64	10.5%
rho_multiplier=2	9.5%	10.0%	0.96	-14.7%	0.64	11.3%

Table 7 shows that increasing the robust penalty tends to raise cash allocation and reduce maximum drawdown, at the cost of lower annualized return in this sample. The case with zero robust penalty is reported only as a sensitivity test, not as a main baseline.

Table 8: Covariance estimator sensitivity. Ledoit-Wolf is a robustness case, not the final implemented risk model.

Estimator	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar	Avg Cash	Real./Pred.
Sample Covariance	9.8%	10.2%	0.97	-16.2%	0.60	10.2%	1.21
Ledoit-Wolf Shrinkage	10.2%	10.4%	0.98	-15.1%	0.67	9.7%	1.31
Final Implemented Covariance Method	9.8%	10.2%	0.96	-15.7%	0.62	9.8%	1.25

Table 8 compares covariance estimators. Ledoit-Wolf shrinkage performs well in this sensitivity table, but it is not the implemented final risk model. The final specification uses the 20-day EWMA covariance estimator because it reflects the chosen short-horizon risk-control design. The comparison also shows that conclusions can be sensitive to covariance estimation, which is an important limitation.

Table 9: Transaction-cost sensitivity for the proposed strategy.

Cost Rate	Ann. Return	Ann. Vol.	Sharpe	Max DD	Ann. Cost Drag	Final NV
0 bps	10.1%	10.2%	1.00	-15.7%	0.00%	4.25
5 bps	10.0%	10.2%	0.98	-15.7%	0.18%	4.14
10 bps	9.8%	10.2%	0.96	-15.7%	0.37%	4.04
25 bps	9.2%	10.2%	0.92	-15.8%	0.91%	3.75

Table 9 confirms that higher transaction costs reduce net performance. Moving from zero costs to 25 bps lowers annualized return from 10.1% to 9.2% and final net value from 4.25 to 3.75. The model remains economically reasonable under the tested cost grid, but a more detailed market-impact model could change this assessment.

Subperiod results further show that performance is regime-dependent. During the 2022 stock-bond drawdown, the proposed strategy loses 9.8% annualized with a -6.5% maximum drawdown, compared with larger drawdowns for Equal Weight and Traditional MVO. During the COVID crash and recovery period, however, the proposed strategy's Sharpe ratio is only 0.21 and its maximum drawdown is -15.7%, illustrating that fast crashes remain difficult for the risk model to avoid.

9 Implementation Checks

Table 10: Constraint, solver, and no-look-ahead implementation checks.

Check	Value
Max budget violation	2.22e-16
Max long-only violation	0.00e+00
Max upper-bound violation	2.26e-10
Max target-volatility violation	4.95e-10
Solver failures	0
Fallback uses	0
Rebalance dates	180
Timing checks passed	5/5

Table 10 reports numerical feasibility and timing checks. The maximum target-volatility violation is below 5×10^{-10}, there are no solver failures or fallback uses, and all five timing checks pass. The timing checks verify that signals and covariances are computed using information available no later than the rebalance date, that weights are applied only after the rebalance date, and that BIL is excluded from alpha and covariance construction.

10 Discussion and Limitations

The results suggest that robust forecast-aware optimization can improve certain risk-adjusted metrics relative to the selected baselines in the tested sample. The framework is interpretable: alpha comes from momentum and trend signals, risk comes from EWMA covariance estimation with PSD repair, robustness is represented by an ellipsoidal alpha-uncertainty penalty, and cash enters through an explicit BIL sleeve. The model is also implementable because it is a convex optimization problem with transparent constraints and explicit transaction-cost accounting.

Several limitations are important. First, expected-return forecasts remain noisy. The signal diagnostics show positive average IC, but also substantial time variation and weaker performance during drawdown periods. Second, covariance forecasts can underestimate realized risk in sudden crises. The realized-to-predicted volatility ratio indicates that the EWMA estimator is not a complete solution to risk-forecast error. Third, the transaction-cost model is simplified. It accounts for full-vector turnover at a constant one-way cost rate, but does not model bid-ask spread differences across ETFs, taxes, market impact, or liquidity-dependent slippage.

The results also depend on the ETF universe, sample period, and fixed parameter choices. Parameters such as α_{scale}, ρ_t, the EWMA decay, and the target-risk bounds are economically motivated but partly heuristic. The paper does not claim that the proposed strategy is optimal in real markets or that it will outperform in future periods. Future work could study alternative covariance forecasting models, richer transaction-cost models, alternative cash proxies, international ETF universes, and formal drawdown-aware optimization as an extension. The final model in this paper, however, does not use any drawdown-based rebalancing trigger.

11 Conclusion

This paper studies a cash-allowed robust forecast-aware optimization framework for multi-asset ETF allocation. The model combines momentum and trend forecasts, annualized 20-day EWMA covariance estimation with PSD repair, a robust forecast uncertainty penalty, a dynamic target-risk budget, full-vector turnover costs, and a BIL cash sleeve. The framework is evaluated through monthly rolling out-of-sample backtests against Equal Weight, Minimum Variance, and Traditional MVO baselines.

In the tested sample, the proposed strategy achieves the highest final net value and the highest Sharpe ratio among the four strategies. It also reduces maximum drawdown relative to Equal Weight and Traditional MVO, though not relative to Minimum Variance. Diagnostic results show that the signal has positive but noisy cross-sectional predictive content, the cash sleeve is used meaningfully, and the risk model can still underestimate realized volatility during abrupt stress events.

Overall, the evidence suggests that robust forecast-aware optimization with an explicit cash sleeve provides a useful and interpretable framework for multi-asset ETF allocation. The results should be interpreted with caution because the model remains exposed to forecast uncertainty, covariance estimation error, transaction cost assumptions, and regime changes.

A Methodology-Code Consistency

Table 11: Methodology-code consistency checklist for the final proposed strategy.

Item	Implemented value
α_{scale}	0.03
EWMA covariance decay	0.94
Transaction cost rate	0.001 per one-way turnover
L2 concentration penalty	0.01
v_t	$\text{clip}(\sigma_t^{EW,EWMA20}/\sigma_t^{EW,252}, 0.5, 2.5)$
ρ_t	$\text{clip}(0.08v_t, 0.03, 0.25)$
σ_t^*	$\text{clip}(0.10/v_t, 0.04, 0.16)$
Single-ETF upper bound	0.25
Cash proxy	BIL
BIL in alpha/covariance	Excluded from both

B References

References

Asness, C. S., Moskowitz, T. J., and Pedersen, L. H. (2013). Value and momentum everywhere. *The Journal of Finance*, 68(3):929–985.

Barroso, P. and Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1):111–120.

Boyd, S., Busseti, E., Diamond, S., Kahn, R. N., Koh, K., Nystrup, P., and Speth, J. (2017). Multi-period trading via convex optimization. *Foundations and Trends in Optimization*, 3(1):1–76.

DeMiguel, V., Garlappi, L., and Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953.

Faber, M. T. (2007). A quantitative approach to tactical asset allocation. *The Journal of Wealth Management*, 9(4):69–79.

Goldfarb, D. and Iyengar, G. (2003). Robust portfolio selection problems. *Mathematics of Operations Research*, 28(1):1–38.

Harvey, C. R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M., and Van Hemert, O. (2018). The impact of volatility targeting. *The Journal of Portfolio Management*, 45(1):14–33.

Jagannathan, R. and Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683.

Ledoit, O. and Wolf, M. (2004). Honey, I shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110–119.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.

Moreira, A. and Muir, T. (2017). Volatility-managed portfolios. *The Journal of Finance*, 72(4):1611–1644.

Moskowitz, T. J., Ooi, Y. H., and Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2):228–250.